

Biomechanical aspects of injury in specific sports

with BEN CLARSEN

Nothing compares to the simple pleasure of riding a bike.

John F. Kennedy (1917–1963)

This chapter outlines the relationship between biomechanics and injury for a number of sports—cycling, cricket fast bowling, golf, rowing, swimming and tennis.

CYCLING

with PHIL BURT

The demands and injury risks differ greatly across the various disciplines of cycling. Traditional road cycling involves long-duration submaximal effort, and places stresses on the body due to monotonous loading and maintenance of static postures for extended periods. In contrast, BMX and track sprint cycling require maximal effort over a short duration. Riders from these disciplines are more likely to suffer injuries related to strength and power training, such as weightlifting and plyometrics. The various subdisciplines of mountain biking encompass a wide range of demands, and both acute and overuse injuries are common. This section focuses on the main overuse injury problems affecting cyclists across all disciplines—knee and low back pain.

Relationship between risk factors and loading

Overuse injuries in cycling are commonly blamed on extrinsic factors such as bike position or shoe and pedal setup, as well as intrinsic factors such as anatomical anomalies, poor cycling technique, or reduced neuromuscular control. These factors may certainly be important due to the repetitive, uniplanar nature of the sport. However, it is important to recognise that even among top professionals, a wide variation of anatomy, techniques, and bike setups is normally well-tolerated without injury occurring. In almost all cases of cycling-related overuse injury, symptom onset can be linked to a

mismanagement of training and racing loads. Cyclists are most likely to develop injury following a rapid increase in load, such as when pre-season training is resumed after a winter break, as well as during intense periods of the season. When injuries are apparently ‘caused’ by a change in equipment, it is normally because the change was made at an inappropriate time of the season when the cyclist was already close to their limit of load tolerance.

The key to successful management of cycling injuries, therefore, is load management. The clinician, cyclist and coach should establish the volume, intensity and frequency of cycling that the rider can tolerate and create a systematic plan to increase these parameters over time. Wherever possible, loading should be quantified using a power meter and training software should be used to monitor the acute and chronic training load during rehabilitation (Chapter 12).

Once an appropriate training plan is established, intrinsic and extrinsic risk factors should be assessed. The following section covers biomechanical factors thought to be associated with the most common cycling injuries. It is important to note that there is little high-level evidence in the field of cycling injuries, with current practice largely based on indirect evidence and expert opinion. It is therefore necessary to take a trial-and-error approach; to be worthwhile, modification in technique or equipment should lead to an obvious improvement in symptoms.

Knee pain

The knee is the most common site of overuse injury among cyclists of all levels.^{1–5} A majority of knee complaints are related to the patellofemoral joint.⁶ However, there is a range of differential diagnoses including iliotibial band syndrome (ITBS), infrapatellar fat pad impingement, medial plica irritations and medial patellofemoral ligament

strains.⁶⁻⁸ Although tendinopathy is generally rare in cyclists, pain can also arise from the quadriceps tendon entheses on the superolateral or superomedial patella.

Three biomechanical factors to assess in patellofemoral pain

Various biomechanical factors may play a role in the development of patellofemoral pain in cyclists, including patellofemoral joint compression forces, knee kinematics in the frontal plane and rotational torques in the lower limb.⁹⁻¹²

1. Saddle height

Patellofemoral joint contact pressure is inversely related to saddle height,⁹ leading to the common belief that cycling with lower bicycle saddle heights increases the risk of patellofemoral pain development.^{8, 11, 13} However, this remains to be confirmed in high-quality risk factor studies of competitive cyclists and one study found that altering the saddle height within the normal ranges used in rehabilitation led to negligible changes in patellofemoral joint contact pressure.¹⁴ Nevertheless, it is advisable that cyclists with patellofemoral pain ride in a relatively high saddle position, with maximal knee extension of approximately 30° (Fig. 9.1).

2. Medial knee motion

Excessive medial motion of the knee in the frontal plane (Fig. 9.2a) may also be a risk factor for patellofemoral pain, as this position encourages lateralisation of the patella



Figure 9.1 Maximum knee extension is a key bike fitting parameter, which typically ranges from 35–40° among professional riders. Patellofemoral joint contact pressures may be minimised by selecting a higher saddle position, with maximum knee extension between 30° and 35°.

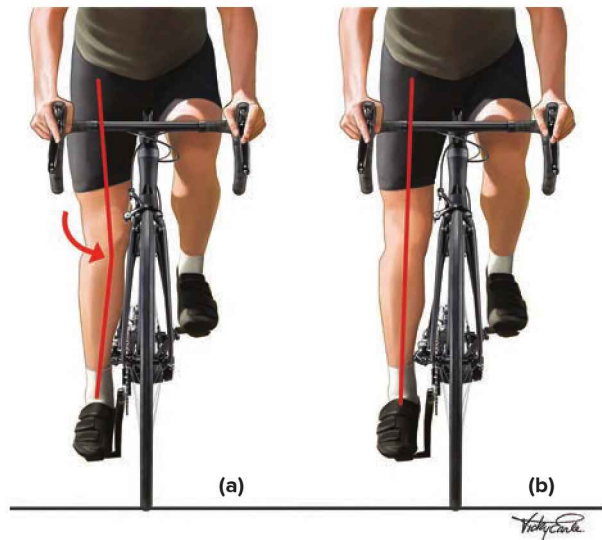


Figure 9.2 Frontal-plane knee motion in cycling
(a) Excessive valgus motion of the knee is thought to contribute to a range of knee injuries (b) Improving the alignment of the hip, knee and ankle may lead to symptomatic improvement

in the femoral trochlea, increased lateral joint contact pressure and increased stress on medial soft tissues. This theory is supported by studies showing that cyclists with a history of knee pain adopt a more medial knee position compared to uninjured cyclists,¹⁰ and that ‘normalising’ frontal plane motion (Fig. 9.2b) can lead to symptomatic improvement.¹⁵ Knee motion may be altered by motor control training or through manipulation of the cyclist’s shoes and pedals. For example, foot position can be adjusted using small angled wedges between the shoe and the pedal cleat or underneath the forefoot or by using custom-made insoles. However, manipulation of cycling shoes and pedals has an extremely unpredictable effect on knee motion.^{16, 17} It is therefore important to test each individual’s response, making sure that adjustments lead to symptomatic improvement.

3. Rotational torque

Rotational torque at the knee caused by the fixation of shoes to the pedals may also be a factor in patellofemoral pain in cycling. After the introduction of modern cleated pedals in the 1980s, there were anecdotal reports of an increase in the prevalence of knee injuries among cyclists.^{6, 12} It was thought that the natural rotation of the lower limb during the pedalling cycle was constrained by fixing the shoe, leading to increased stress at the knee joint. Therefore, ‘floating’ pedals were designed that allowed a small degree of axial rotation, which attenuated the rotational torque at the knee.¹² Although there is no

direct evidence that floating pedals reduce injury, their design has been widely accepted and they remain the most popular type used by cyclists today.¹

Iliotibial band syndrome

ITBS, often called 'runner's knee', is also common among cyclists.^{6, 18} Intrinsic factors thought to contribute to ITBS include large leg length discrepancies, external tibial torsion of greater than 20° and excessive subtalar pronation.¹⁸ High saddle height, a *toe-in*, *heel-out* foot position and excessive medial knee motion also increase the stress on the ITB.¹⁹

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Therefore, cyclists with lateral knee pain should be instructed to ride with a relatively low saddle (maximum knee extension approximately 40°), keep their feet straight or pointed slightly outwards and avoid a knee-in position.

Low back pain

Although transient back discomfort is normal in cycling, studies have shown that performance-limiting low back pain is common among amateur²⁰ and elite cyclists.^{1, 21}

Cyclists with low back pain typically present with non-specific symptoms provoked by the maintenance of sustained flexion positions and they can often be classified as having a flexion-pattern motor control dysfunction (Chapter 29).²² Using a remote posture monitoring system, Van Hoof et al. showed that cyclists with low back pain

adopt a more flexed position in their lumbar spine than do pain-free cyclists (Fig. 9.3).²³ This may be related to a number of pathomechanical mechanisms of low back pain,^{22, 24} such as flexion/relaxation inhibition, or fatigue of the erector spinae muscles and mechanical creep of the spine's viscoelastic tissues. However, these theories remain largely untested in cyclists.

Encouraging a relaxed, anteriorly tilted pelvic position, with an even distribution of flexion throughout the spine, is often important in the overall management of cyclists with low back pain. A number of equipment modifications may help facilitate this, including lowering the saddle, raising the handlebars and shortening or lengthening the overall reach.

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Adjusting the saddle to a slightly nose-down position (2–4°) may also be helpful in achieving greater anterior pelvic tilt and reducing back pain.²⁵

Excessive lateral flexion and/or rotation of the spine while cycling may also contribute to back pain, particularly if it is asymmetrical. This can be caused by a range of factors, such as large leg length differences, hip range of motion limitations and asymmetrical muscle activation patterns. These should be considered as a part of the comprehensive management of the cyclist with low back pain. Bike fit is a critical element in such a comprehensive assessment (see box).

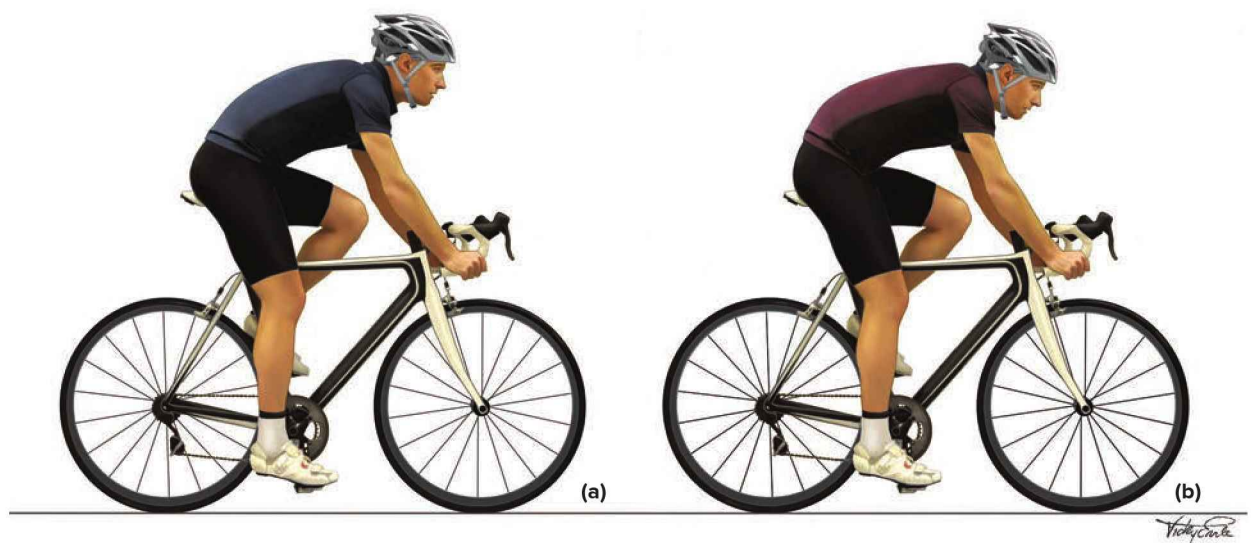


Figure 9.3 Spinal position in cycling (a) Ideally, forward bend should be achieved evenly (b) Cyclists with low back pain often adopt a more flexed lumbar spine, with less anterior pelvic tilt and a more extended thoracic spine

Bike fitting

There are many approaches to bike fitting, ranging from simple anthropometry-based formulae to dynamic approaches utilising high-tech equipment (Table 9.1 and Fig 9.4).

As bike fitting involves the optimisation of a wide range of competing variables, such as aerodynamics, comfort and control, it always involves compromise. Despite the recent rapid pace of technological development in the cycling

industry, there remains little research into bicycle equipment and injury, and bike fitting remains just as much an art as it is a science. We encourage clinicians to work closely with bike fitters who take a trial-and-error approach, considering the cyclist's previous and current injuries, cycling goals and physical limitations.

Table 9.1 Advantages and limitations of modern bike fitting methods

Formula-based approach	
Description	There are a number of formula-based approaches that convert anthropometric measurements (e.g. inseam height) to bicycle set-up parameters. The most well-known approach is named after two-time Tour de France winner, Greg LeMond.
Advantages	Quick and easy. Cyclists can perform measurements themselves.
Limitations	One-size-fits-all approach that does not consider the cyclist's physical limitations. Highly unreliable and variable results. ²⁶
Static angle-based approach	
Description	The cyclist's major joint angles are measured with a goniometer while he or she sits on the bike without cycling. The bike is adjusted to position each joint within a predetermined 'optimal' range of motion. Sometimes referred to as the Holmes method.
Advantages	Good reliability. ²⁷
Limitations	Does not consider dynamic cycling technique. 'Optimal' angles are not evidence-based.
Dynamic angle-based approach	
Description	Dynamic measure of joint angles collected with two-dimensional or three-dimensional motion analysis over a period of time and averaged as rider is actually riding.
Advantages	Good reliability. ²⁷ Accounts for the rider's technique and physical limitations.
Limitations	Does not consider kinetic variables such as power distribution between legs.
Combined-input approach	
Description	Combination of data streams such as power, pedal forces, saddle pressure with motion analysis.
Advantages	Multiple sources of data can lead to more-informed clinical-reasoned decisions.
Limitations	More data does not always lead to better clinical decisions. Little research to assist data interpretation. Validity and reliability unknown.

Figure 9.4 Modern bike fitting is an interactive process incorporating data from a variety of sources such as three-dimensional motion analysis, shoe and saddle pressure distribution, and pedal force application

IMAGE COURTESY OF CYCLEFIT UK



CRICKET FAST BOWLING

with ALEX KOUNTOURIS

Cricket fast bowling combines rapid near-end-range trunk movements (rotation, side flexion and extension), with high vertical and horizontal impact forces.²⁸⁻³⁰ Elite fast bowlers bowl in excess of 300 deliveries in a single 4-5 day game, and have a higher rate of injury than any other players in cricket.³¹

Soft tissue and bone injuries are typically reported in injury surveillance studies of fast bowlers.³² Lumbar spine injuries have received the most attention mainly because bone stress injuries of the posterior vertebral elements (pedicle, pars interarticularis and lamina) require lengthy recovery periods and are particularly common.³¹

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The most important risk factors are bowling biomechanics, age and workload.

Research into cricket fast bowling injuries has identified a number of possible modifiable and non-modifiable risk factors, particularly for lumbar spine injuries.^{32, 33} It is likely that a combination of these factors determines the injury risk for fast bowlers, although a clear association between the three factors has not been established.

Age and risk of bowling injury

To highlight this important interplay between risk factors, younger fast bowlers, for example, are affected by biomechanical (technique) errors because they have an immature skeletal system that cannot cope with the high forces associated with fast bowling. As such, younger (adolescent) fast bowlers have a higher incidence of lumbar bone stress injuries than adult bowlers and have a disproportional high rate of lumbar spine radiological abnormalities and soft tissue injuries for their age.^{34,35} Technique (biomechanical) faults that increase the amount of trunk rotation, extension and side flexion increase the load on the already susceptible musculoskeletal structures like the intervertebral disc and posterior vertebral arch.

Bowling workload and injury

Bowling workload is a modifiable factor associated with fast bowling injuries in both adolescent and adult fast bowlers. High bowling workloads in a single session, in a match or over a cricket season, have caused injuries in fast bowlers.^{29, 36-38} More importantly, workload 'spikes' (a rapid increase in workload) have also been associated with fast bowling injuries and particularly lumbar spine injury.^{39, 40}

The relationship between bowling technique and workload is not entirely clear, but it is likely that bowlers with a technique that is considered to be biomechanically sound may be able to tolerate higher workloads than those with other 'less safe' techniques because they are better equipped to dissipate forces. This relationship could be further complicated when considering age and that some structures may not have the resilience to absorb repetitive load (e.g. the posterior vertebral arch) until full maturity is reached. For example, in the lumbar spine, important geometric parameters that influence bone strength are not fully developed until the mid-twenties. In long bones, increases in bone mineral density (BMD) and bone mineral content (BMC) occur between pre-puberty and 18 years of age, but in the lumbar spine the increases in BMD, BMC, vertebral height and vertebral width continue to develop until 25 years of age.⁴¹ This age (25 years) seems to be an important time in the development of lumbar bone stress and other lumbar spine injuries.

The relative impact of workload and age as risk factors needs to be considered along with bowling technique, because technical faults that result in exaggerated spinal positions and higher impact forces can magnify the impact of other risk factors.

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Biomechanical errors are arguably the most important risk factor for fast bowling injury.

Like other sporting activities, fast bowling technical faults are considered modifiable factors although making meaningful technical changes can be difficult.⁴² To fully understand the biomechanical faults it is important to review the patterns of movement (kinematics) and forces (kinetics) that occur during the fast bowling action.

Fast bowling technique and injury

The simplest description of the fast bowling action is that it involves a combination of trunk rotation, and lateral flexion and extension during the delivery stride.²⁸⁻³⁰ The sequence of movements and the associated ground reaction forces (GRF) during the fast bowling delivery stride are unique to every bowler, but can be broadly categorised into one of four bowling techniques: front-on, side-on, semi-open (semi-on) and mixed (Fig. 9.6).⁴³ These bowling techniques are defined by position of the feet, pelvis, trunk and shoulders at specific times during the delivery stride, and assessed using video (two-dimensional) or ideally, three-dimensional computer analysis.⁴³

The fast bowling action can be broken down into three components that are interrelated and

important for both performance and injury.⁴⁴ The main components are the run-up, the delivery stride and follow-through. The run-up is important as it allows bowlers to gain the momentum required to deliver the ball at a desired velocity and can pre-set the body position that occurs during the delivery stride. The delivery stride is the last ground contact of the back foot, known as back foot contact (BFC), and then the transition to front foot contact (FFC) and ends with ball release (Fig. 9.5). The highest GRF, between six to nine times body weight, occur during the period between FFC and ball release.^{28, 29, 45} The follow-through begins at ball release and involves deceleration of the bowler until they come to a stop.

Importance of biomechanics of the delivery stride

Bowlers can be categorised into the four bowling techniques based on the positions and movements that occur during the delivery stride.⁴⁴ The four main bowling techniques are outlined in Figure 9.6.

Of the four bowling actions, the mixed technique has been identified as having the greatest association with lumbar spine injury and abnormal radiological features (Fig 9.6d).^{35, 46} The maximum amount of shoulder-pelvis separation angle occurs just after FFC,⁴³ and possibly represents the period that places the bowler at greatest risk of injury, especially as it is also the time when the greatest ground impact forces are being absorbed.^{29, 45} The mixed technique also typically involves higher rates of trunk contralateral side flexion and hyperextension compared to other bowling actions and therefore places greater stress through the lumbar spine.⁴⁷

Another biomechanical aspect that has been associated with lumbar spine injuries in fast bowlers is shoulder counter-rotation (SCR), which is also closely associated with the mixed bowling technique. Bowlers with higher rates (greater than 30°) of SCR during the delivery stride have been associated with higher incidence of lumbar spine bone stress injuries and abnormal radiological features.^{28, 29, 35, 44, 48} SCR (Fig. 9.7) is the change in shoulder alignment between BFC and FFC, from a more front-on to a more side-on alignment.^{43, 44} The measurement of SCR is made using both two-dimensional and three-dimensional motion analysis and can vary slightly depending on the definition of BFC (first-foot contact with the ground versus foot flat on the ground).⁴³

It is still unclear whether one particular bowling trunk position causes the most stress on the lumbar spine or if a combination of movements is required to exceed load tolerance. Cadaver studies indicated that lumbar extension was most likely to be associated

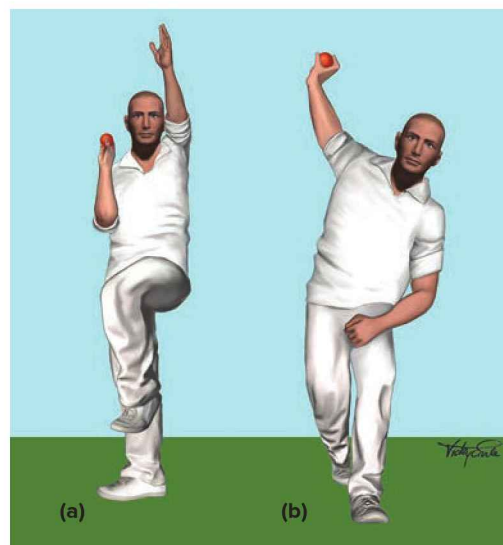


Figure 9.5 Cricket fast bowling components (a) Back foot contact (BFC) (b) Front foot contact (FFC)

DRAWING ADAPTED FROM PORTUS ET AL.⁴⁴

with lumbar bone stress injury in fast bowlers, because of increased load on the posterior vertebral arch in a hyperextended position.^{28, 49} Since then, the role of trunk side flexion and rotation has also received attention. Ranson et al. proposed that the amount of contralateral side flexion and ipsilateral rotation at FFC were important factors in the development of lumbar spine injury in fast bowlers.⁵⁰ They found that the greatest range of motion in the lower lumbar spine at FFC was contralateral lumbar side flexion and not lumbar extension as first thought. They concluded that the combination of high rates of contralateral side flexion, ipsilateral rotation and the large impact forces would be associated with lumbar spine bone stress in fast bowlers.⁵⁰ It is most likely that a combination of trunk positions is responsible for the high force required to develop lumbar spine injuries, in particular, bone stress injuries and the individual contribution of each movement to injury risk could be associated with the bowling technique used.

Apart from the spine and trunk positions adopted during the delivery stride, lower limb joint kinematics have also been linked with fast bowling injuries. The amount of knee extension at FFC influences the distribution of forces and influences the height at which the ball is released, so that a higher release height requires less knee flexion (straighter knee).^{29, 51} The increase in front foot knee extension leads to a stiffer knee segment and greater impact forces.^{44, 51}

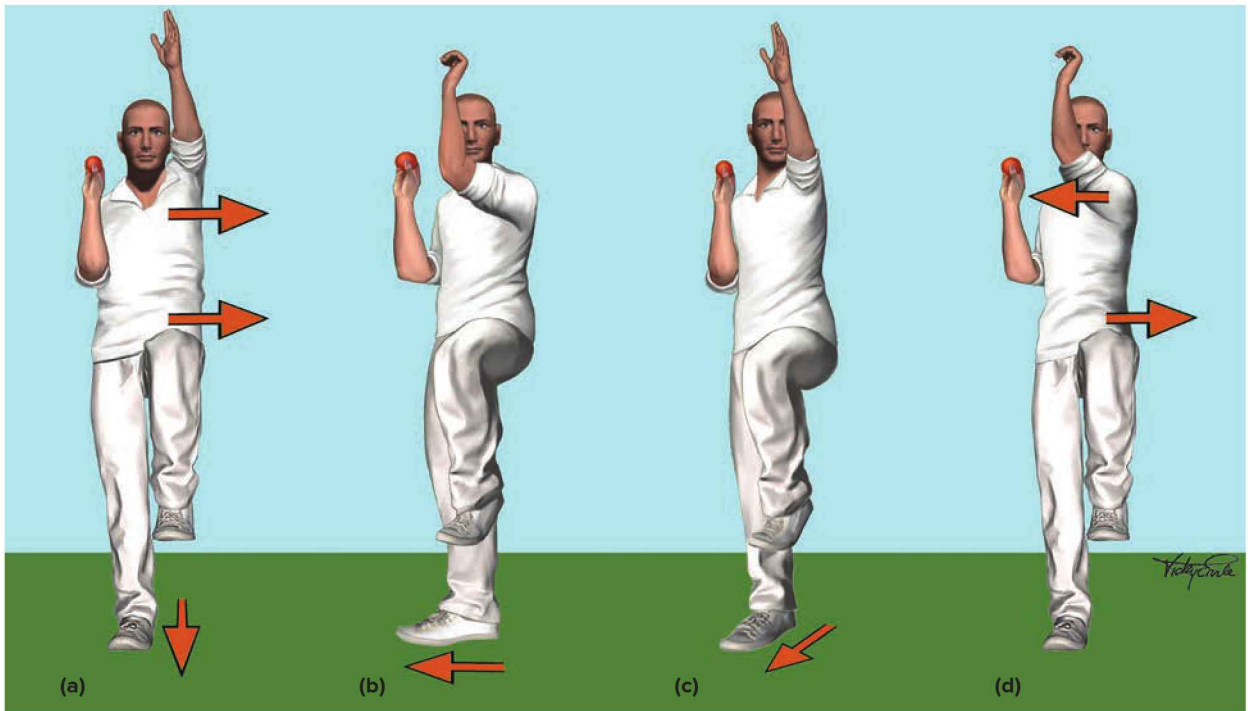


Figure 9.6 Cricket fast bowling techniques

(a) *Front-on technique* involves the alignment of the pelvis and shoulders parallel to the batsman at back foot contact (BFC)

(b) *Side-on technique* involves the alignment of the bowler's shoulders and pelvis perpendicular to the batsman at BFC

(c) *Semi-on or semi-open technique* is defined as both shoulder and pelvis alignment somewhere between front-on and side-on positions at BFC

(d) *Mixed technique* involves dissociation between the shoulder and pelvis alignment at BFC, known as the shoulder–pelvis separation angle. For example, a front-on pelvis alignment and side-on shoulder alignment at BFC, or side-on pelvis position and front-on shoulder position at BFC. A shoulder–pelvis separation angle of 30° has been used to classify bowlers as having a mixed technique

DRAWING ADAPTED FROM PORTUS ET AL.⁴⁴

Portus et al.⁴⁴ reported a relationship between a more extended front knee at FFC and ball release, and higher braking and vertical impact forces. They also demonstrated that bowlers with greater knee extension also reached peak forces more quickly than those with a more flexed knee at ball release. The higher forces associated with a more extended front knee may increase injury risk but also allow bowlers to deliver the ball at faster bowling speeds.⁴⁴ This is an important performance benefit and needs to be considered if technique modification is contemplated. It is therefore possible that there may be a trade-off between faster bowling speeds and higher injury risk.

In summary, it is important to consider bowling biomechanics as a critical risk factor for fast bowling injuries. The relative impact of biomechanical errors on injury risk should be considered along with bowling workload and the age of the fast bowler, as there is a delicate interplay between these key risk factors.

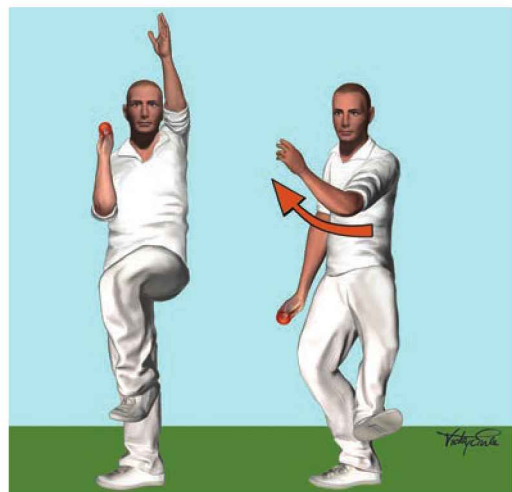


Figure 9.7 Shoulder counter-rotation during delivery stride

DRAWING ADAPTED FROM PORTUS ET AL.⁴⁴

GOLF

with ROGER HAWKES

Although injury rates are generally low, golf-related injuries are still common because of the popularity of the sport. It can be played into old age where the health and social benefits lead to increased longevity.⁵²

Golf relies on a coordinated and repeated action in which energy is generated in the lower body and trunk. In long hitters, the force is transferred efficiently into the upper limbs and ultimately the club head, which can travel at speeds of 190 km/h. Professional tournaments are usually comprised of four rounds of 18 holes over a 4 day period during which the player also includes considerable practice. It is estimated that the average tour professional will hit around 200 shots per day. In addition, there has been an emphasis on hitting the ball as far as possible. It is not surprising therefore that, in spite of improving equipment and better conditioning, vulnerable tissues remain susceptible to injury. For instance, peroneal tendons in the leading ankle are used during the follow-through to prevent excessive inversion. Overuse can lead to tenosynovitis and lateral ankle pain and swelling. In addition, any deficiency in the kinetic chain may lead to compensation elsewhere and injury is more likely to occur. For instance, a player who is not able to rotate their thoracolumbar spine during the backswing may compensate by increasing the horizontal adduction of the leading shoulder in order to reach the top of the backswing. This may lead to shoulder injury.⁵³ Furthermore, fatigued players will have a reduced club head speed but an increased range of movement and this adds to injury risk.⁵⁴

This chapter focuses on wrist, shoulder, hip and low back pain in golfers. Medial elbow pain, which is also common, is covered in Chapter 25.

Wrist pain

Good amateurs and professional players require considerable practice and intentionally take divots in order to impart backspin on the ball. In the winter they often practice on mats. As a result, the leading wrist can be painful.⁵⁵ This is particularly common in periods of the year when the ground is hard.

The leading wrist

The leading wrist is the more vulnerable of the two wrists during the golf swing because it moves from radial deviation at the top of the backswing (Fig. 9.8a) to ulnar deviation during follow-through. The trailing wrist follows a less stressful extension/flexion movement pattern (Fig. 9.8b). At impact, the ulnar side of the leading wrist is protected by the extensor carpi ulnaris

tendon, which is susceptible to overuse synovitis. When a player hits an obstruction, the impact can rupture the sub-sheath which holds its position at the lower end of the ulna. This structure is particularly vulnerable in supination, which occurs during the golf swing and results in a large angle between the muscle axis and the tendon as it inserts into the fifth metacarpal.⁵⁶

As a result of repetitive movement into ulnar deviation during the backswing, the leading wrist is also susceptible



(a)



(b)

Figure 9.8 (a) The leading wrist (left) is in maximal radial deviation at the top of the backswing (shown) and moves into ulnar deviation during the follow-through (b) The trailing wrist (right) is in maximal extension at the top of the backswing (shown) and moves into flexion during the follow-through

to bone stress reactions, particularly of the hamate. Older players may develop osteoarthritic changes in the carpus, and players with long ulnas (either constitutionally or acquired after radial fracture) may be susceptible to ulnar-sided wrist pain as a result of impaction.

Fractures of the hook of hamate are rare, but 35% of them occur in golfers.¹¹ The hook of hamate is particularly vulnerable when a tall player uses standard length clubs. As a result, the end of the club may press onto the hook of hamate or piso-hamate ligament. Acute fractures can occur if the club hits the ground or an object, and with repetitive impacts the hook is also susceptible to bone stress injuries.

The trailing wrist

The trailing wrist extends during the backswing (Fig. 9.8b) and flexes during the follow-through. Players with dorsal rim impaction syndrome⁵⁷ pinpoint pain at the radial carpal joint in loaded extension, but often have no pain in passive extension. It is usually thought to occur as a result of nipping of the synovium, which can become chronically inflamed and hypertrophied. In some cases it can be due to loose bodies. Ultrasound is a good starting investigation but can be normal. Magnetic resonance imaging (MRI) can also show hypertrophy of the synovium (Fig. 9.9) and computed tomography (CT) scans can show loose bodies.

Players with tenosynovitis will need to reduce practice in order to reduce their symptoms. Amateur players may also choose a more forgiving shaft such as those made of graphite. They may improve the sweet spot by using hollow back clubs rather than blades. Other general treatments are covered in Chapter 26. Players who acutely sublux their extensor carpi ulnaris (ECU) tendons



Figure 9.9 Dorsal rim impaction syndrome—Sagittal MR image showing synovial/capsular hypertrophy (arrows) of the dorsal aspect of the radiocarpal synovium. Normal capsule would be a thin low signal line

generally prefer to have operative repair of the subsheath, but conservative treatments have been used in tennis players with good results with no apparent increased time away from their sport. Some players have asymptomatic subluxing ECU tendons and do not require treatment.⁵⁵

Players with dorsal rim impaction syndrome respond well to corticosteroid injection. However, the condition can return and arthroscopic resection of the thickened synovium can be indicated.

Finally, players who have had hook of hamate stress reactions or fractures will need to use longer clubs. If there is non-union of a fractured hook of hamate, resection is usually preferred in order to reduce the risk of injury to the ulnar nerve and artery injury, which can complicate pinning.⁵⁸

Shoulder pain

The forces acting on the leading shoulder in golf are almost completely opposite to those involved in throwing (Chapter 8). The shoulder hyperadducts during the backswing, which can over stretch the posterior structures. During the follow-through, anterior structures work eccentrically to slow the shoulder down. The resultant combination of anterior capsular tightness and posterior laxity pushes the humeral head backwards, placing stress on the posterior labrum which can lead to posterior instability.⁵³ Players typically complain of pain or clunking at the top of the backswing. Clinical examination may show signs of posterior instability and MRI may show injury to the posterior labrum.

Players with a history of anterior instability in the leading shoulder may develop pain and apprehension toward the end of the follow-through when the shoulder is abducted and moving into external rotation. In the trailing shoulder, the movement (and injury) pattern is more similar to throwing sports.

Acromioclavicular joint pathology may occur in golfers. It is aggravated by joint compression, which occurs during the backswing for the leading shoulder and during the follow-through for the trailing shoulder.

In recent years there has been considerable interest in the phenomenon of glenohumeral internal rotation deficit (GIRD) in throwing athletes (Chapters 8 and 24). In golfers, however, the main focus is on identifying and addressing deficits in external rotation. Professional players are regularly screened for glenohumeral external rotation deficits (GERD) in an attempt to prevent posterior instability in the leading shoulder, although there is currently little research to show this as an effective prevention strategy.

Players with shoulder pathologies often need to limit golf participation in the early phases of rehabilitation. Resumption of practice should begin with putting and small chips with high irons, before gradually increasing the swing and playing with progressively longer clubs.

Hip pain

A recent survey of professional golfers found that 19% of players complained of current, significant hip and groin pain.⁵⁹ Common hip diagnoses in golfers include labrum pathology/femoroacetabular impingement and, in older players, osteoarthritis.

A majority of golf-related hip problems are located in the leading hip,⁵⁹ which undergoes higher rotational velocities and moves through a much greater range of motion than the trailing hip.^{60, 61} However, both hips need to be able to move freely to facilitate an effective swing—particularly into internal rotation. The trailing hip goes into internal rotation at the top of the backswing and the leading hip is at maximum internal rotation at the end of the follow-through. Players with osteoarthritis and femoroacetabular impingement may present with ‘hip pain’ at either end of the swing.

Occasionally, players will present with reduced internal rotation in the leading hip without hip joint pathology. This may be due to tightness of the posterior structures that work eccentrically during the follow-through. Limited internal rotation of the leading hip can have a range of consequences on swing mechanics, such as causing the player to ‘stand up’ during the follow-through. Maximising hip internal rotation range of motion is therefore important for all golfers.

The effective internal rotation range of motion may also be increased by encouraging players to take a more externally rotated stance. This is recommended for players with femoroacetabular impingement and osteoarthritis, as well as for players who have had hip replacement surgery.

Low back pain

Younger players may be vulnerable to spondylolysis, especially if they end their swing with a ‘reverse C’ shaped thoracolumbar spine (Fig. 9.10). In addition, recent trends towards a more aggressive swing involving a larger separation between the hips and the shoulder may also lead to back injury. Both these technique factors should be reviewed in players with low back pain.

Reduced hip internal rotation in the leading hip increases stress on the thoracolumbar spine, as rotational demands of the follow-through are transferred higher up the kinetic chain. Tight hip flexors are also a risk factor because this leads to more anterior pelvic tilt and therefore more ‘jamming’ of the facets in the lumbar spine.⁶²

ROWING

with FIONA WILSON

The biomechanics of rowing are complex with many variables contributing to the speed of the boat. Each of

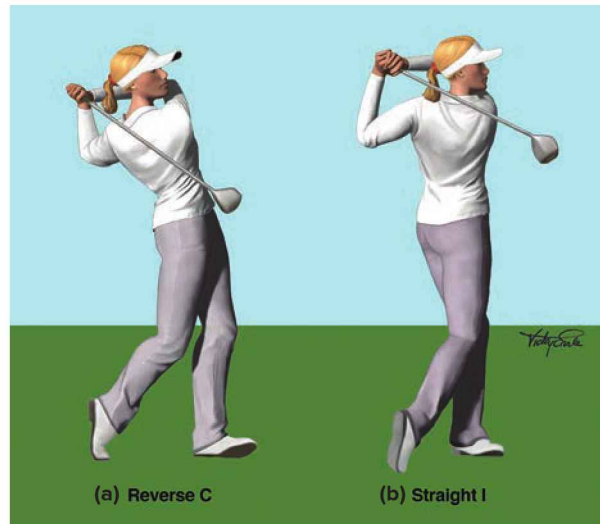


Figure 9.10 Follow-through position (a) Reverse C position increases stress on the lumbar spine, predisposing the player to spondylolysis (b) Ideal follow-through position for the lumbar spine. Note the large range of hip internal rotation in the leading hip

the three body segments contributes approximately one third of the stroke length. The legs execute their work when the force exertion is maximal in the first half of the stroke and produce nearly half the rowing power, with the trunk producing around one third and the arms only one fifth of the power output. Thus, contrary to common belief, the legs are the greatest source of power in rowing, increasing their contribution as the stroke rate increases.⁶³ High forces are generated at several specific points of loading on the rower's body. Due to the cyclical movement pattern and high volume of training, these forces are repeated hundreds of times during a typical training session. It is this combination of high forces acting on the rower, large training volume and type of training that places the rower at risk of injury.

This section will focus on three of the most common regions of rowing-related pain—the lower back, chest wall and wrist/forearm.

Low back pain

Between 32% and 50% of rowers experience rowing-related low back pain per year,^{64, 65} and there are a number of biomechanical factors thought to contribute to the development of low back pain. The spine is maximally loaded at the front of the stroke—known as the ‘catch’ position—where the blade of the oar is placed in the water and force is applied through the trunk, arms and legs.

Ideally, forward reach at the catch is achieved through maximal hip, knee and ankle flexion, with the lumbar spine in slight flexion and the pelvis in anterior rotation (Fig. 9.11a). However, inadequate lumbopelvic control or limitations in hip, knee or ankle range of motion may lead to a position of extreme lumbar flexion at the catch, with the pelvis in posterior rotation (Fig. 9.11b). In this position the intervertebral discs and posterior elements of the spine may be more vulnerable to injury.⁶⁶⁻⁶⁸

Good lumbopelvic control is also important in the finish position. Here, a neutral lumbar spine is ideal (Fig. 9.12a), whereas collapsing into lumbar flexion increases stress on the spine (Fig. 9.12b). Some rowers may also have inadequate posterior rotation of the pelvis, leading to hyperextension of the lumbar spine in the finish position.

Rowing technique deteriorates during continuous rowing leading to increased lumbar flexion and frontal



(a)

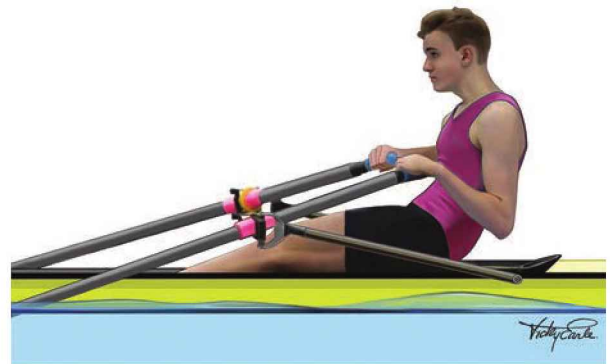


(b)

Figure 9.11 The catch position (a) Pelvis in relative anterior rotation with an even distribution of flexion throughout the spine (b) Pelvis in relative posterior rotation increases the stress on the lumbar spine



(a)



(b)

Figure 9.12 The finish position (a) Lumbar spine in a neutral position (b) Lumbar spine collapsed into flexion

plane motion, which is attributed to fatigue.⁶⁹⁻⁷³ However, this could be dependent on rowing ability and experience. Novice rowers use high levels of lumbar flexion with limited pelvic rotation, deteriorating further with higher work intensities.⁷⁴ While similar changes are seen in elite rowers, these are of a much lower magnitude.⁷⁵ A number of factors have been identified as a predictor of lumbar spine injury in rowers. As with most cases of low back pain, a previous episode increases risk.⁷⁶ This is important to note at pre-season screening and those considering entry to the sport. Another factor is time spent ergometer training. Studies have shown that lumbar kinematics may be different in ergometer rowing compared to the boat, specifically that the increased lumbar flexion associated with prolonged rowing and fatigue is of greater magnitude in ergometer rowing.⁷² Therefore, rowers with lower back pain should reduce their exposure to the ergometer and pay close attention to their lumbopelvic control while

rowing, ideally using some form of biofeedback (mirror, video or electrogoniometer).

A majority of rowers with low back pain present with flexion-pattern motor control dysfunction.⁷⁷ As for other patient groups, a multi-dimensional cognitive functional approach (Chapter 28) has been shown to have the best efficacy in rowers.^{78, 79} However, a number of specific considerations should be applied to the rowing population. An exercise approach, which focuses on optimal position of the trunk and pelvis throughout the stroke, should underpin rehabilitation. Excellent range of motion, particularly at the hips, knees and ankles, should be emphasised.

A mistake made by many rehabilitating low back pain in rowing is poor specificity in exercise prescription, particularly overemphasis of isometric exercise in the trunk such as the 'plank' exercise. Rowing is a dynamic sport and rehabilitation exercises aimed at improving trunk endurance should allow fluid movement. The ability to hold static positions affords limited benefits.

Chest wall pain

Chest wall pain is particularly common in elite rowers,⁸⁰ but may occur in rowers of all levels. Most often it is caused by rib stress fractures (RSF), in particular in the anterolateral portion of the fourth to eighth ribs.

Risk factors for rib stress fracture

The mechanism of this injury is not well understood, but it is likely to include mechanical loading of the chest wall as well as individual intrinsic factors such as gender and bone mineral density.⁸¹ A combination of force vectors acting on the ribs from scapula retractors, external obliques and rectus abdominis may initiate a stress response leading to RSF. Training volume (particularly a sharp load increase) and equipment issues such as blade size and type have also been implicated.⁸² Thoracic hypomobility, particularly extension, is common in those sustaining RSF.

The incidence is higher in females and may be hormonally driven or may be associated with upper body strength differences from male rowers. As with any stress fractures, low energy availability is implicated in RSF pathogenesis. This may be particularly relevant in lightweight rowers. Emerging research has noted increased risk of RSF in paralympic rowing athletes.⁸³ This may reflect low thoracic bone mineral density in spinal-cord-injured athletes because of a reduced skeletal loading.

Clinical features

Rowers with RSF typically present with pain in the anterolateral chest wall that is initially aggravated by rowing. As the condition worsens, other activities become painful, point tenderness develops, and pain is aggravated by deep breathing, coughing, rolling over in bed and rising

from supine. Examination may reveal a positive rib spring and reproduction of pain during movements mimicking the rowing stroke,⁸¹ particularly on isometric resistance. Diagnosis is primarily from symptoms and examination and may be confirmed with a bone scan or MRI.

Management

Management is challenging and relative rest is the primary approach for 2–6 weeks. A symptom-dependent approach should be taken and the athlete should make a graduated return to training depending on pain. Many rowers report ongoing symptoms for many months (sometimes intermittently for longer) despite achieving good function. During rest, non-rowing exercise such as cycling can be maintained. Biomechanical factors such as poor thoracic extension or altered movement control should be corrected. Manual therapy of the thoracic spine can be very useful, although compression of the rib cage should be avoided. Taping for proprioceptive feedback may aid comfort. Liaison with the coach and video analysis can be very useful in identifying correctable faults such as 'over-reaching' or poor scapula control at the catch position.

The medical staff of the Great Britain rowing team have published clinical guidelines for the diagnosis and management of rib stress injuries in rowers.^{84, 85}

Wrist and forearm pain

Common wrist and forearm injuries in rowers include intersection syndrome (also known as 'oarsman's wrist'), de Quervain's tenosynovitis and exertional compartment syndrome (Chapters 25 and 26). All three conditions are related to excessive use of the forearm musculature, which is influenced by a variety of factors. Rowing requires the athlete to rotate the oar handle after taking it out of the water ('feathering the blade') and again prior to re-entering the water ('squaring the blade'). Ideally, the blade should be feathered and squared using a combination of wrist and metacarpophalangeal joint movement. However, a common technique fault, particularly among novice rowers, is to use wrist motion only during this action.

High grip pressure is also linked with forearm pain. This is a particular problem when rowing in poor weather conditions and among beginners, who tend to grip the oar more firmly. Oar handle size and grip should be considered. Modern oar handles have rubber grips and may be custom-fitted in different sizes; however, non-elite rowers are more likely to use older oars with wooden and 'one-size-fits-all' handles.

Sweep-oar rowers (i.e. those who row with only one oar), are more likely to suffer from intersection syndrome and de Quervain's tenosynovitis on their inside arm as that wrist is in a greater degree of ulnar deviation. Changing

sides of the boat may help alleviate symptoms and allow for greater participation during rehabilitation.

Knee pain

Rowing-specific knee pain is rare. However, lateral knee pain may occur in rowers who steer coxless boats using a rotating shoe that is connected to a rudder wire. Tendinopathy of the distal biceps femoris tendon may also occur in rowers who actively use knee flexion to 'pull up the slide' rather than swing over from the hips with a stable trunk.

SWIMMING

with ELSBETH VAN DORSEN

The clinical management of swimmers is a challenge. Swimmers begin their career earlier than many other athletes—typically between 8 and 12 years—and often train twice a day for a total of 3 to 6 hours. As elite swimmers perform a staggering 1.5 million strokes per arm every year, overuse injuries are common.

Swimming biomechanics

Swimmers use both the upper limbs and the lower limbs, but in freestyle, backstroke and butterfly the arms generate approximately 80% of the propulsion. Swimmers try to generate propulsive force while reducing the resistance to forward motion. Drag forces (friction drag, form drag and wave drag) are extremely important at all swimming speeds; when you double the speed in water, you quadruple the form drag force. Achieving a hydrodynamically efficient swimming position (long and lean body position, aligned with the direction of travel), while still being able to generate propulsive forces, requires specific strengths, flexibilities and skills. Swimmers will try to increase the length of the propulsive phase (when the arm is in the water), combined with a fast but 'relaxed' recovery phase (above the water).

Swimmers compete in four strokes: freestyle, butterfly, backstroke and breaststroke. However, irrespective of a swimmer's preferred stroke, more than 50% of their training will be spent doing freestyle.⁸⁶

Freestyle technique

The freestyle stroke can be divided into four phases (Fig. 9.13):

1. hand entry
2. reach/glide
3. pull-through
4. recovery phase.

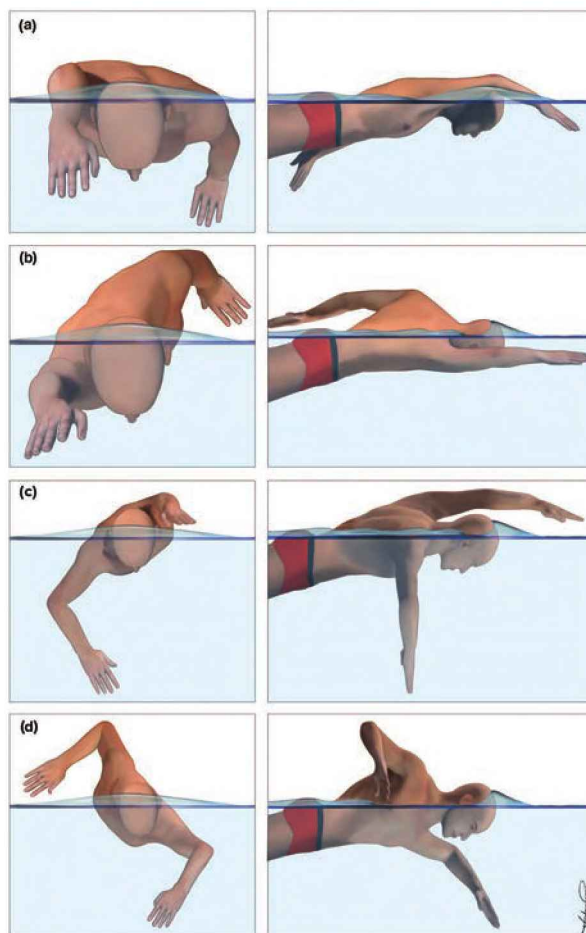


Figure 9.13 Phases of freestyle swimming (a) Hand entry (b) Reach/glide (c) Pull-through (d) Recovery

Shoulder pain

The shoulder region accounts for 31–39% of all swimming injuries.^{87, 88} The prevalence of shoulder pain in elite swimmers has been found to be as high as 91%,⁸⁶ with 10–31% forced to stop training for some time during the year due to shoulder pain.^{87–91}

There is a higher risk for shoulder injuries:

- after an abrupt increase of training volume or intensity⁹²
- with training volume >35 km (or 15 h) per week⁸⁶
- when a unilateral breathing pattern is used⁹³
- after recent change of stroke technique; ask about any technical flaws that have been pointed out by the coach⁹⁴
- when crossing the midline with the hand during pull-through (Fig. 9.13)
- with a history of shoulder injuries⁹⁰
- with a recent change in coach (and therefore likely changed training loads)

Shoulder pain in swimmers—diagnostic labels

The term 'swimmer's shoulder' was introduced by Kennedy and Hawkins.⁹⁶ This is, however, a nondescript and confusing catch-all which does not advance our understanding. We believe the term should be replaced by an individualised and more specific diagnosis, which accounts for the contributing factors (extrinsic and intrinsic) and suspected pathology of each injured swimmer.

Instead we propose **Supraspinatus tendinopathy** with:

- *hypomobility of the cervicothoracic junction*
- *loss of internal rotation of the glenohumeral joint*
- *greater internal rotation strength due to sudden increase of intensity of training and possible overload in the use of hand paddles and drag suits.*

This will promote a patient-specific, problem-oriented approach to treatment.

- after increased use of hand paddles^{89, 95}
- with the use of drag-increasing training devices (bags, elastic cords, drag suits, etc.).

Rotator cuff tendinopathy is the most common shoulder pathology seen in swimmers.⁸⁶ Tendinopathy of the long head of the biceps and impingement against the anterior third of the coracoacromial arch are also frequent diagnoses. However, a clear pathoanatomical diagnosis is often difficult to achieve; it is more important to diagnose and understand the underlying functional or pathomechanical reasons for the 'structural failure' and address these in the management plan.

A thorough clinical examination of the entire kinetic chain is important to assess the intrinsic contributing factors. Swimmers must be capable of extreme ranges of motion, especially in the shoulder girdle,⁹³ and they must be able to reach these positions easily. Mechanical cost to achieve this position will result in (i) unnecessary extra movements and drag, and (ii) suboptimal use and possible overload of musculoskeletal structures. Modelling studies suggest that, on average, the supraspinatus tendon is already in a position of potential mechanical impingement for nearly 25% of the freestyle arm stroke cycle.⁹³

If swimmers are not flexible enough in the glenohumeral joint, they tend to pass this motion requirement to the scapulothoracic joint or the spine. On the other hand, hypermobility can also increase the potential for injury due to excessive translation of the humeral head. There might be a range of 'optimal motion' in swimmers. One study found that those with more than 100° or less than

93° of external rotation had a greater risk of shoulder impingement symptoms.⁹⁰ Internal rotation is the most important, as it allows the swimmer to have an early catch and high elbow. A measure between 40–50° seems to be necessary for freestyle, backstroke and butterfly. Breaststroke swimmers can have a little less.

Testing abduction with internal rotation gives important information about the swimmer's ability to achieve and maintain a high elbow. The swimmer sits on a bench, both arms are passively abducted, with the elbows maintained in 90° flexion (Fig. 9.14). The forearm must be perpendicular to the plane of abduction (causing internal rotation). The angle measured is the line of the humerus to the straight spine. An appropriate range for this measure is between 150° and 170°.⁹⁷

The combined elevation test can give more information about the ability to achieve a high elbow position at the start of the stroke, recovery and streamline. The athlete lies in a prone position with both arms (with locked thumbs together and extended elbows) elevated above the head. The swimmer then elevates the arms as high as



Figure 9.14 The abduction with internal rotation test.⁹⁷

The elbows should be kept in 90° flexion, and the upper arms should be prevented from moving forward as the shoulders are elevated. Swimmers should aim for 150–170° of abduction.

possible while keeping the head, chest and legs in contact with the bench (Fig. 9.15). An angle between the line of the humerus and the horizontal axis of between 5° and 15° is appropriate.⁹⁷

To maintain a streamlined position, core stability is important. Swimmers need to transfer load efficiently between the arms and the legs under internally generated flexion torques.

Management of shoulder pain in swimmers

Individualised rehabilitation programs emphasising range of motion, flexibility, muscle balance, and motor control of the glenohumeral and scapulothoracic joints are the key features in the management of shoulder pain in swimmers (Chapter 24).⁹⁸ Modification of training load is often necessary.

Most commonly, the entire kinetic chain is involved, with multiple intrinsic and extrinsic contributing factors such as:

- reduced rotation and/or extension of the thoracic spine—important in relation to body roll
- tightness or improper motor control around the scapula
- muscular imbalance around the shoulder, in particular, internal and external rotators
- incorrect swimming technique (Table 9.2)
- too much training with hand paddles or drag-increasing devices
- improper land training, weight training or stretching
- suboptimal core strength and/or control.

A team approach: technical stroke analysis and adjustment with the coach, physiotherapist, doctor and sports scientists, keeping the clinical findings in mind.



Figure 9.15 The combined elevation test—a test for thoracic extension, shoulder elevation and scapular retraction.⁹⁷ The swimmer lies in a prone position and is asked to lock their thumbs together, extend their elbows and elevate the arms as high as they can while keeping their head, chest and legs in contact with the bench. The humerus angle is measured relative to horizontal

Table 9.2 Technique factors related to shoulder pain in freestyle swimming

Presentation	Possible cause	Solution
Shoulder pain at hand entry	Thumb-first hand entry—excessive internal rotation in full elevation	• Palm-down hand entry • Ensure adequate abduction/internal rotation range of motion
	Crossing the midline with the hand at hand entry	• Avoid crossing the midline
	Lateral deviation of the trunk	• Improve mobility in the kinetic chain, improve core stability
Shoulder pain during pull-through	Insufficient upward rotation of the scapula	• Address scapulohumeral muscle flexibility • Train scapular upward rotation in overhead exercises
	Dropped elbow—elbow-first water entry or elbow below the level of shoulder and wrist during early pull-through	• Improve internal rotation ROM—sleeper's, stretch and cross-body stretch—check kinetic chain
	Crossing the midline with the hand during pull-through	• Avoid crossing the midline
Shoulder pain during recovery	Insufficient or excessive body roll	• Body roll should be approximately 45°; this can be limited by thoracic rotation and core stability
	Excessive horizontal abduction due to insufficient body roll—posterior cuff impingement	• Increase body roll • Optimise mobility of cervicothoracic spine
	Unilateral breathing pattern (pain on same side)	• Strive for a bilateral breathing pattern

Swimmers with shoulder complaints should avoid the following drills:

- long leg drills with a kick board, as arms are maintained in maximum shoulder flexion; this can aggravate impingement symptoms
- hand paddles, as these increase the load on the shoulder during the pull-through
- pull buoy; the legs are held together without kicking, so 100% of the propulsion is generated by the arms.

Medial knee pain

Medial knee pain is common in breaststroke swimmers.⁹⁹ In one study, 75% of the surveyed breaststroke swimmers reported at least three episodes of knee pain per season and 47% of these swimmers reported weekly episodes of knee pain.¹⁰⁰

Although the term 'breastroker's knee' is nonspecific, it usually refers to overuse injury of the medial collateral ligament. However, the medial patellofemoral joint, pes anserine tendon, or bursa and adductor muscles may also be involved.¹⁰¹

The medial knee is highly loaded at the beginning of the breaststroke thrust phase, where the knee is flexed and externally rotated and the hips are flexed and internally rotated. The knee undergoes valgus loading throughout the leg thrust. A number of intrinsic factors and technique flaws can increase the valgus load and contribute to medial knee pain:

- valgus alignment of the lower limb
- inadequate or asymmetrical hip internal rotation or tibial external rotation
- large abduction angle of the hips during the kick (>30°)
- ineffective arm-leg coordination may increase loads on the lower limb
- sudden increase in amount or intensity of training
- hip extension and ability to achieve core and pelvic stability.

Clinical assessment of the lower back, hips, knees, and ankles is important, particularly the range of hip internal rotation and tibial external rotation. These two measures should add up to approximately 90°, so that at the beginning of the thrust phase the feet are square to the line of progression. This results in the biggest area to use to propel in the kick with the knees closer together, thus reducing form drag.⁹⁷

As for shoulder injuries, the clinician should focus on both intrinsic and extrinsic factors during the rehabilitation process. Breaststroke training should be reduced, and exercises which place a valgus load on the knee should be avoided during land and weight training. Hip rotation range of motion should be optimised and the load capacity of the

knee should be increased through progressive lower limb strengthening.

TENNIS

with BABETTE PLUIM

In a typical hour of tennis, players hit the ball between 150 and 250 times and run between 2.5 and 3.5 km in high-intensity 3-metre bursts. Due to the game's explosive, intermittent and repetitive nature, both acute and overuse injuries are common. Groups particularly at risk include elite players, who often train up to 5–6 hours per day, deconditioned players who suddenly increase their training loads and young players who play more than four matches per week.¹⁰²

The main biomechanics-related tennis injuries are lateral elbow pain ('tennis elbow') and overuse shoulder injuries. These represent approximately 15–25% of all tennis injuries.¹⁰³

Lateral elbow pain

Lateral elbow tendinopathy is colloquially referred to as 'tennis elbow' as it has been estimated to affect up to 50% of active tennis players at some point throughout their career.^{104, 105} The prevalence among tennis players seems to be declining, which may be related to increasing use of the two-handed backhand across all levels of tennis. The condition is caused by a tendinopathy of the common extensor tendon, in particular the extensor carpi radialis brevis, at the insertion on the lateral humeral epicondyle. Here, forces are greatest during one-handed backhand shots, as the wrist extensors contract eccentrically on impact with the ball.

Lateral elbow pain is much more common in beginners and players over 40 years of age,¹⁰⁶ and is usually attributed to a combination of overuse and poor stroke biomechanics. Skilled players hit the ball with an extended wrist and extend the hand through impact (Fig. 9.16a), whereas novice players often hit the backhand with a flexed wrist, resulting in an eccentric contraction and less strength at impact (Fig. 9.16b).^{107, 108} However, the relationship between this technique flaw and elbow injury remains unclear,¹⁰⁹ and other potential factors include hitting the ball off-centre and squeezing the grip too tightly during follow-through.¹¹⁰ The topspin backhand usually provokes more complaints than the slice backhand.¹¹¹

PRACTICE PEARL

Tennis elbow is very rare in players with a two-handed backhand.

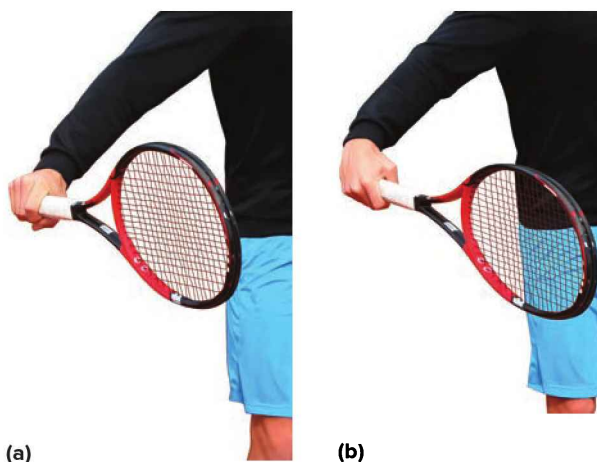


Figure 9.16 One-handed backhand technique (a) An extended wrist at impact reduces stress at the lateral epicondyle (b) A flexed wrist at impact increases stress at the lateral epicondyle

Tennis racquet grip size is traditionally considered to be a risk factor for lateral elbow pain, as grip sizes that are too small or too large may lead to extra strain on the forearm muscles. It is therefore important to optimise the racquet grip size to the player.¹¹² Other common recommendations to reduce load on the forearm muscles include using a larger racquet head, a flexible racquet, lower string tension (with high quality strings) and pressurised tennis balls. Playing with heavy, wet tennis balls, long hours of play, or a sudden increase in volume or intensity of play may lead to sudden overuse and the development of lateral elbow problems.

The general management principles for lateral elbow tendinopathy are detailed in Chapter 25. However, tennis-specific management should include developing muscular strength to better cope with the racquet-ball impact, and an analysis of backhand technique, with effort focused on using the right grip and achieving concentric wrist extension at the point of impact. Slow-motion video analysis may be a useful biofeedback tool to help players change technique. Low compression balls should also be considered during the early phase of treatment.

Shoulder injuries

The shoulder is subjected to high loads in tennis, particularly during the serve where the shoulder is abducted to 140–160° and rotates 160–180° at a rate of up to 2400°

per second.¹¹³ As a player serves approximately 100 times in an average match,¹¹⁴ overuse shoulder injuries are common. Both the causes and the pathology of shoulder injuries in tennis are similar to other overhead and throwing sports (Chapter 8); common diagnoses include internal impingement, rotator cuff and biceps tendinopathy and glenoid labrum injuries.^{115, 116}

During the cocking phase of the serve, the supraspinatus, infraspinatus and serratus anterior all contract concentrically to stabilise the scapula and glenohumeral joint. This is followed by a rapid eccentric contraction of the rotator cuff and serratus anterior to decelerate the arm during the follow-through. These repetitive concentric and eccentric demands, which occur during both serving and groundstrokes, may lead to muscular fatigue and so-called 'eccentric failure' of the tendons. Adding further stress, the supraspinatus and infraspinatus tendons can become compressed between the humeral head and the posterosuperior rim of the glenoid during the cocking phase of the serve, when the glenohumeral joint is in maximal external rotation and horizontal abduction. This is often referred to as *internal impingement*.

Local factors that contribute to internal impingement include muscular imbalances (external rotator weakness or dysfunction compared to the internal rotators), weakness or dysfunction of the scapular stabilisers (scapular dyskinesis) and tightness of the posterior capsule resulting in reduced internal rotation range of motion (GIRD). These factors should all be considered when treating shoulder injuries in tennis (Chapter 24).

Remote factors also play an important role in the development of shoulder injuries in tennis. To generate power during the serve, there should be coordinated activation of a number of body segments (legs, hips, trunk, shoulder, arm and hand)—the so-called 'kinetic chain'—to achieve a high racquet speed at impact.¹¹⁷ Suboptimal kinetic chain mechanics may overload the upper limb as it tries to make up for lost energy production.¹¹⁸ This is referred to as the 'catch-up phenomenon'.¹¹⁶

Common clues that point to poor kinetic chain mechanics during serving (Fig. 9.17) include:¹¹⁹

- insufficient knee bend—the front knee should flex more than 15° during the loading phase
- inadequate rotation—the player should be coiled enough that they show their back to the opponent
- inadequate shoulder and pelvis tilt during the loading phase.



Figure 9.17 Tennis serve technique—loading phase. Look for a deep knee bend, rotated trunk, and tilted shoulders and pelvis



REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e